

YOURSTORY



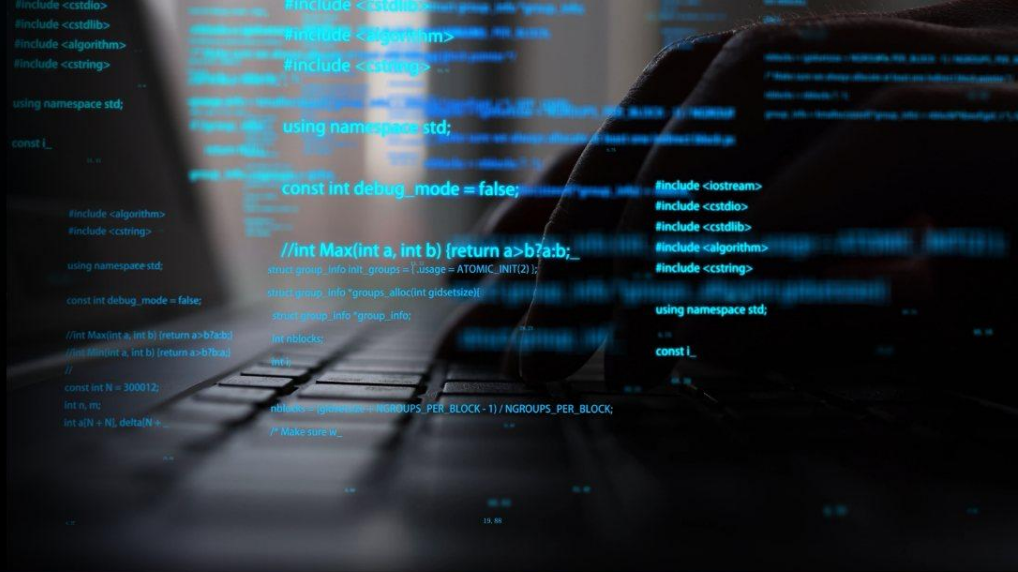
H2S



Coco CLI

HACKATHON

GCC Edition



Team Name : **Workforce Astra**

Team Leader Name : **Rahul Rao**

Team Size : **1**

Problem Statement : **Customer 360 and Next Best Action Engine (as Employee 360)**

1 · PROBLEM BRIEF

Three teams, three headcounts, from the same data

THE BUSINESS PROBLEM

HR, Finance and Engineering each re-derive headcount, attrition and pay metrics ad hoc. Nobody owns the definition, so the same question gets a different answer from each team.

Ask “how many employees do we have?” and get 109 from one query, 127 from another, on identical data.

PAIN POINT → IMPROVEMENT

Today: reconciliation before every board pack; pay-equity numbers nobody trusts; exit interviews read once and never joined to anything.

With Workforce Astra: one owned, tested definition per metric; interview reasons joined to pay and performance; the next action drafted for a human to approve.

WHO IT'S FOR

- People Analytics and HR business partners
- Finance: workforce cost and board reporting
- Engineering and delivery leaders: spans, retention
- CHRO and compliance: pay equity, band policy

INDUSTRY CONTEXT: GCCs

A Global Capability Centre runs people operations for a global parent. The parent, the centre and its vendors each keep their own definitions, and contractors blur what “headcount” means.

Customer 360 unifies the customer. Employee 360 unifies the workforce.

SOLUTION · EMPLOYEE 360

One governed definition, so every tool gives the same answer

SAME DATA, SAME MOMENT, TWO ANSWERS

109

Governed FTE headcount, from the Employee 360 semantic view

127

Ungoverned “active” count: silently includes 18 contractors

The semantic view makes the governed answer the default, in every tool.

NEXT BEST ACTION

An MCP tool drafts the manager alert, comp adjustment or band review. Every tool defaults to dry-run: nothing is sent, and a human approves.

Five governed Semantic Views

Employee 360 headcount, attrition and comp-ratio across HRIS and reviews

Org Health 360 span of control, overspan managers, org depth

Pay Equity 360 raw vs band-adjusted pay gap; small cohorts suppressed

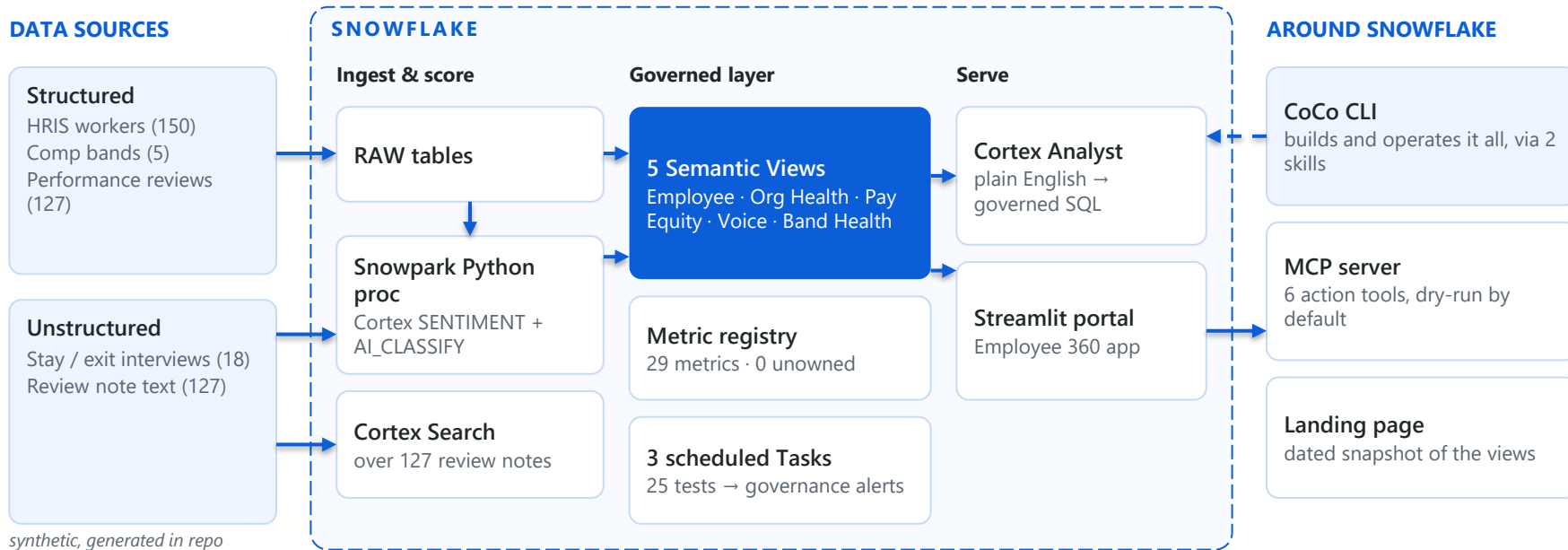
Employee Voice 360 interview sentiment and reason, joined to risk

Band Health 360 below-minimum pay, red circles, band overlap

2 · ARCHITECTURE

Data flow: sources to governed views to action

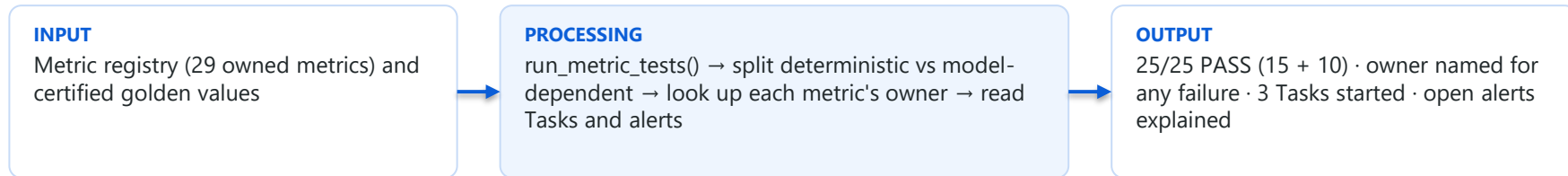
DATA SOURCES



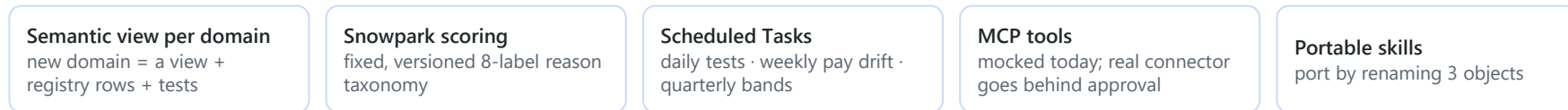
CoCo skill 1 `$workforce-astra-data-gen` loads the data · **CoCo skill 2** `$workforce-astra-governance-check` runs 25 tests, names owners

2 · ARCHITECTURE: COCO CLI SKILLS

Two reusable CoCo skills, and modules that plug in

Skill 1 `$workforce-astra-data-gen`Skill 2 `$workforce-astra-governance-check`

HOW MODULES PLUG TOGETHER



MVP RESULTS · SYNTHETIC DATA, LIVE SNOWFLAKE

What the governed views find

PAY EQUITY

-2.8%

band-adjusted gender pay gap (raw: -16.8%)

- Board and legal read one governed number
- 39 of 44 cohorts suppressed (under 5 people)

BAND HEALTH

37 of 150

paid below their own band minimum (24.7%)

- 0 red circles: the obvious metric misses it
- M1 and IC5 ranges overlap 71%

EMPLOYEE VOICE

5

corroborated flight risks: negative, underpaid, rated 4+

- 18 interviews scored in Snowpark + Cortex
- Top reason: career growth (6 of 18)

ORG HEALTH

9.0

average span of control across 15 managers

- 5 managers over 10 direct reports
- Reorg what-if tool is guarded

EMP-0031 exit interview · sentiment -0.53 · comp-ratio 0.80 · rated 4 of 5

"I found out in a meeting, by accident, that a colleague who joined after me on the same band is earning materially more."

The structured data already said flight risk; only the transcript says why. 5 of the 18 synthetic interviews were placed on the risk profile deliberately, so the join has something to find.

GOVERNANCE & GUARDRAILS

Trust is built in: owned, tested, guarded

29

metrics in the registry

0 unowned: every metric has a named owner and a certified definition

25/25

golden-value tests pass

15 deterministic · 10 pin Cortex model output

3

scheduled Tasks

daily regression · weekly pay drift · quarterly bands

Model changes can't slip through. Ten tests pin Cortex output, so a model upgrade fails loudly instead of quietly moving a board number.

Small cohorts stay private. Pay figures for cohorts under 5 people are withheld automatically; pay columns are masked.

No automated pay decisions. The band auditor only reports: no proposed-pay column, and the MCP tool can't be asked for one.

Actions are drafts. All 6 MCP tools default to `dry_run = true`. Nothing reaches Slack or Workday.

No real people. Every record and transcript is synthetic and labelled as such in the data.

3 · IMPACT STATEMENT

One number per metric, and actions you can defend

MEASURABLE IN THE MVP

- Headcount overstatement caught: 127 vs 109 (+16.5%), resolved to one governed answer
- Pay gap explained: 14 points separate the raw –16.8% from the adjusted –2.8%
- 37 below-minimum employees surfaced that the red-circle metric (0) misses
- 5 at-risk strong performers flagged, with the reason in their own words
- 25 automated checks replace manual metric reconciliation

SCALABILITY

- Views, registry and tests don't change as headcount grows; the warehouse scales
- A new domain is a view + registry rows + tests, proven five times here
- Portal compute suspends when idle; one-command rebuild on a fresh account
- The CoCo skills port to another project by renaming 3 objects

BEYOND THE DEMO

- Real Workday and Slack connectors, behind human approval
- More domains: learning, hiring funnel, internal mobility
- Row-level security by manager hierarchy
- A Marketplace starter kit: synthetic data + semantic views
- Fits any GCC reconciling parent, centre and vendor definitions

To measure in a pilot: time to reconcile board metrics, and time from risk signal to manager action. MVP figures above are on synthetic data.

BUILT WITH

Snowflake-native, built and run through CoCo CLI

Semantic Views

5 governed domains

Cortex Analyst

plain English → SQL, 0 warnings

Cortex Search

127 review notes

Cortex AI SQL

SENTIMENT, AI_CLASSIFY

Snowpark Python

interview scoring procedure

Streamlit in Snowflake

Employee 360 portal

Tasks

3 scheduled governance jobs

CoCo CLI

2 skills; every DDL applied through it

MCP server

6 tools, mocked, dry-run

Verified live on 26-Sep-2026: a full CoCo CLI run loaded the data, passed 25/25 tests, returned 109 vs 127 and answered through Cortex Analyst.

LINKS

Demo video

[\[add unlisted YouTube link\]](#)

Code

github.com/rahulrao85/Workforce-Astra

Live page

workforce-astra.rahulrao.in**DATASETS**

All data is synthetic, generated by scripts in the repo; no real person or employer. Sources and licences: README → Datasets and licences.



YOURSTORY



1125

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THANK YOU